

ParkProof

Build Journal — the 10 days that shipped it

A chronological record of how ParkProof went from an empty repo on 16 May 2026 to a launch-ready, custom-domain, internationalised, cloud-synced PWA — 100+ commits across 10 active build days, shipping for a Reddit launch on Tuesday 26 May. Each day gets a narrative, the commits that landed, and a decision-tree block at every pivot where a chosen path replaced one I'd ruled out.

Read this as a build artefact, not a sales document. It's honest about the dead ends — including the ~5 hours on 20 May spent trying to bypass the API Gateway 30-second timeout with two architectures that failed before the third one stuck, and the silent PDF-export bug caught on 24 May during a pre-launch audit that would have shipped to roughly a third of supported users.



The arc, at a glance

Date	Theme	Headline work
16 May	Foundation	Initial commit + reminders + screenshot harness + PDF fixes
17 May	The Big Push	Live countdown · Cognito + DDB + S3 cloud sync · 5-language i18n · Layer 2 telemetry
18 May	Polish + Launch	Custom domain · 16-screenshot pipeline · case study · UX bug-fix sweep
19 May	Safety Gates	Pre-launch audit · paid-parking acknowledgement · no-sign-here flow
20 May	Crisis + Pivot	Accessibility gate · 30s timeout crisis → async-polling architecture · end-of-session feature
21 May	Harden + Domain	Test coverage filled (66 cases) · lambda contract tests (20) · 2 prompt fixes · DNS migration · www.parkproof.com.au live
22 May	De-identify + Push	Apple Bundle ID rotation · web push end-to-end · 7 → 9 languages · visual pop refresh
23 May	Two-App Cutover	Marketing landing at / · React PWA at /app/ · considered-and-deferred docs
24 May	Pre-launch Hardening	PDF font crisis caught · Lambda warmer · /verify page · reminder visibility surface · 18 more

25 May

Launch Eve

Sign-in CTA gap fix · screenshot refresh · Reddit thumbnail polish ·
case-study tightening

DAY 1 · Fri 16 May 2026 · Foundation

Day one is an initial-commit day. The repo was created from a working state — Sign Translator, Session Logger, basic reminders, PDF export, session history all scaffolded — and the first commit is essentially the entire app in one push: 88 files, ~17,400 lines. That's the snapshot of a previous build session getting pushed up to GitHub.

The three follow-up commits are the kind of work you do once the bones are in place: deepen the reminder picker (multi-offset with auto-disabled past chips), build a Playwright screenshot harness so the README can carry real captures, and harden the PDF export against the edge cases that broke when I tried to render sessions that had no signature yet.

Commits

98bd332	chore: initial commit — 88 files, ~17,400 lines
e6396b4	feat(reminders): multi-select offsets with auto-disabled past chips
5a236b4	feat(tooling): automated screenshot harness via Playwright
98d2f17	fix(pdf): surface export errors, guard malformed signatures, fix expired-session copy

DECISION · Where to render the parking-spot timezone — device or spot?

Q: Show '3:45 pm' in the user's device locale, or in the parking spot's local time?

```
+-- A · Device locale
|   (simpler; works without GPS)
|   ✗ Travelling Sydneysider scans in Melbourne, sees device-Sydney
|     time on a Melbourne sign. Off by an hour at DST boundaries.
|
CHOICE -+
|
+-- B · Spot timezone via tz-lookup ← CHOSEN
|   ✓ Times always match the sign's reality
|   ✓ One npm dep (tz-lookup, 6.1 MB) vs hand-rolled offset math
|   ✓ Works offline (lookup is bundled, not API)
```

DAY 2 · Sat 17 May 2026 · The Big Push

The single heaviest day of the build — 15 commits including the two largest single landings of the project: a 3,495-line cloud-sync pivot (Cognito User Pool + JWT-gated API Gateway routes + DynamoDB + S3 evidence bucket, end-to-end) and a 1,704-line internationalisation expansion (Hindi + Punjabi added on top of the five-language scaffolding from earlier the same day).

The order matters: Tier-A features (photo-quality pre-check, walk-back navigation, the home 'Currently parked' card with urgency-coloured countdown, KMS cryptographic signing) shipped *before* the auth pivot. That kept the user-facing surface complete-as-a-free-app first; cloud was layered on as opt-in durability second. Anonymous-by-default is a deliberate ParkProof design value — every feature works without a login wall.

DECISION · Cloud-sync backend — which managed identity + storage stack?

Q: How do users sign in optionally, and where does their data live?

```

      +-- Firebase Auth + Firestore
      |   X New cloud vendor, separate billing,
      |   another security surface to harden
      |
      +-- Supabase (Postgres + Auth + Storage)
      |   X Tempting (great DX), but introduces a
      |   second hyperscaler runtime in the stack
      |
Identity + storage  -----+
      |
      +-- AWS Cognito + DDB + S3 ← CHOSEN
          ✓ Same AWS account as the Lambda
          ✓ Cognito JWT authoriser bolts onto the
            existing API Gateway routes natively
          ✓ KMS signing key already lives here
          ✓ One bill, one IAM model, one CloudWatch
          ~ Cognito's Hosted UI is dated – accept it
            for federated login, custom-style email/pw
```

DECISION · Adding multi-lingual UI — locale picking strategy?

Q: Which languages to ship and how to source the list defensibly?

```

      +-- Most-spoken globally (CN, ES, AR, FR, RU...)
      |   X Generic. Doesn't match the user.
      |
      +-- Most-spoken in Australia (incl. EN regional)
      |   X Wide list, low signal per locale added.
      |
Locale set  -----+
      |
      +-- City of Melbourne LGA top non-English ← CHOSEN
          langs from 2021 ABS Census
          ✓ Direct match to actual user population
          ✓ Final 7: EN, Zhongwen (zh-CN), Tieng Viet,
            Italiano, Ellinika (el), Hindi (hi), Punjabi (pa)
          ✓ Defensible in any PM interview as a
            data-grounded scoping decision
```

Commits

a58cd4c	feat(home): live 'Currently parked' card with urgency-coloured countdown
9c0691b	feat(time): date-aware 'Until' / 'Move by' / reminder labels
402f531	feat+fix: Tier-A features + three gotcha fixes (1,053+ lines)
1d20352	docs(readme): refresh stale sections after Tier-A + gotcha shipments
c393511	feat(auth+sync): Cognito sign-in, DynamoDB + S3 cloud sync, end-to-end (3,495+ lines)
c8e7391	fix(auth): SPA boot polyfill, CORS for GET + Authorization, export as PDF
a5cd755	docs(readme): align with the now-shipped auth + PDF-export reality
3127e86	docs(readme): flag pending Apple federation
84c7d99	feat(i18n): five-language support with flag selector — EN / zh-CN / vi / it / el
d5f6f3b	docs(readme): align with multi-lingual + auth + cloud-sync reality
9542071	fix(i18n): restore inline emphasis lost when strings were flattened
d3d0419	fix: ReuseCard Italian half-translation + stale-chunk PDF export crash
6e84cff	feat(i18n): Hindi + Punjabi + PDF translations + dropdown selector (1,704+ lines)
2cdc257	docs(readme): Apple federation live
03f8006	feat(telemetry): AI feedback Layer 2 — context-aware failure analytics

DAY 3 · Sun 18 May 2026 · Polish + Launch Prep

The shape of the day changes — the deploy pipeline is humming, the feature set is broadly complete, and the brief shifts from *build features* to *make this defensible as a portfolio piece*. Twelve commits, three parallel threads: domain cutover, screenshot pipeline rebuild, and the PM case-study doc.

Bug-fix interludes throughout: the "+N more" pill on the active card was visually a tap target but technically wasn't (button-inside-button is invalid HTML and silently swallows the click); home-screen cards drifted off-centre when wrapped i18n strings pushed the layout; cross-device image sync wasn't actually working when users re-opened on a new device — root cause was S3 presign URLs minted under the wrong path prefix.

DECISION · Custom domain — registrar route?

Q: parkproof.dsouza.tech is on Network Solutions. How do we point it at CloudFront with a valid TLS cert?

```
+-- Transfer registrar to Route 53
|   X ~7-day transfer window, $ cost,
|   breaks the existing dsouza.tech setup
|
+-- Use Route 53 as DNS-only (NS records
|   delegated from Network Solutions)
|   X Adds a second DNS provider to maintain,
|   Network Solutions' NS-delegation UI is
|   known to be quirky
|
Domain → CloudFront -----+
|
+-- ACM cert in us-east-1 + CNAME on      ← CHOSEN
    Network Solutions pointing at the
    CloudFront distribution
    ✓ Single CNAME row; Network Solutions
      panel is enough
    ✓ ACM auto-renewal stays in AWS
    ✓ Cert + distro live in the same account
    X ~2 hours lost to Network Solutions
      DNS UI quirks (slow propagation,
      no DNS preview)
```

Commits

3ab2e6f	feat(domain): cut over to parkproof.dsouza.tech
79d5acc	chore(public): drop unused / off-brand orphan assets
fb1fa8b	docs: refresh README + CLAUDE for current shipped surface
7bbe469	chore: add MIT LICENSE
34cef80	docs(readme): fix stale numbers + add cloud-sync to feature list
8b6d103	fix(i18n): home-screen cards drift off-centre when translations wrap
5116c7e	feat(screenshots): expand pipeline to 16 captures + add real fixtures
558aed1	docs(screenshots): regenerate all 16 from the expanded pipeline

73df423	docs(readme): restructure Demo into 4 themed sub-grids
e8388b9	docs: add PM case study (docs/case-study.md)
b82eef6	fix(home): "+N more" pill is now a real tap target, not a UX lie
e3e1ceb	fix(sync): cross-device images now actually load

DAY 4 · Mon 19 May 2026 · Safety Gates

A quiet day by commit count (2) but big by impact. The first commit is the pre-launch audit — fixes for stale doc references, missing alt-text, OG-image metadata. The second is the day's substantive feature: **paid-parking + no-sign-here**. Both are safety mechanisms.

The paid-parking acknowledgement is the most under-appreciated decision I made on this project. Without it, ParkProof says 'you can park now' for a paid bay without the user explicitly acknowledging they need to pay — which is the exact failure mode that creates a wrongful-feeling ticket. The gate turns an implicit assumption into an explicit choice.

DECISION · Detecting 'no sign here' parking — how to capture the spot defensibly?

Q: User wants to log a park at a spot with no posted restrictions — what evidence do we capture?

```

      +-- Trust the user; save GPS + arrived_at only
      |   X Weak evidence if a council later disputes
      |       and claims a sign was always there
      |
      +-- Force the user to scan something anyway
      |   X Defeats the purpose; the spot has no sign
      |
No-sign capture  ----+
      |
      +-- Capture an *ambient surroundings* photo ← CHOSEN
          (substitutes for sign photo as visual evidence
           of 'no posted restrictions at this spot')
      + GPS + reverse-geocode + arrived_at
      ✓ Tamper-proof photo evidence
      ✓ No AI call — no token spend, no error path
      ✓ Renders in evidence PDF with a distinct
        'No posted restrictions' banner
      X Open-ended (no expiry) — defers the
        'how do these sessions ever end?' question
        to a later day (it became Day 5's feature)
```

Commits

9fb2d45	docs: pre-launch audit fixes
11dcad4	feat: paid-parking gate + no-sign-here flow

DAY 5 · Tue 20 May 2026 · Crisis + Architecture Pivot

Today started normally and turned into a full architecture pivot. Morning: the accessibility-permit gate shipped (a parallel to yesterday's paid-parking gate — a hard RED warning when the sign requires a disability permit, blocking the Save button on an explicit acknowledgement checkbox).

Then a LinkedIn commenter publicly reported a sign that returned `{"message":"Service Unavailable"}`. The sign — a stacked Pratt Street monster (Clearway + multi-arrow + accessibility + meter zones) — was taking Claude 30-50 seconds to read carefully. API Gateway HTTP API has a hard 30-second timeout. The first commit (84ccbb1) was a band-aid: catch the 5xx, retry once, surface a friendly 'try a clearer / cropped photo' message instead of the raw AWS error.

But complex signs are exactly the kind ParkProof is supposed to help with. The band-aid was unacceptable. I spent the afternoon attempting two architectures that should have worked — both failed in ways I couldn't reasonably solve in the session window — and then pivoted to a third architecture that did work.

DECISION · Bypassing the API Gateway 30s timeout — three attempts

Q: Complex signs take 30-50s. API Gateway HTTP API caps at 30s. How?

```
+-- A · CloudFront → Lambda Function URL (OAC, sigv4)
|   Idea: CloudFront has a 60s origin-response timeout.
|       OAC signs requests to a Function URL with
|       AuthType=AWS_IAM, bypassing the gateway.
|   X Persistent sigv4 signature mismatches.
|       CloudFront's signing payload didn't match what
|       the Lambda Function URL was verifying – multiple
|       AWS forum threads describe the same dead-end.
|       Burned ~90 min. Reverted.
|
+-- B · Cognito Identity Pool + anonymous sigv4
|   Idea: Frontend gets anon AWS creds, signs Lambda
|       Function URL requests directly. No gateway.
|   X 403 from the Function URL despite IAM-simulate
|       showing the role had iam:lambda:InvokeFunctionUrl.
|       Same-account principal worked; the Cognito role's
|       signed requests didn't. Some account-level block I
|       couldn't identify. Burned ~60 min. Reverted.
|
30s ceiling --+
|
+-- C · Async-polling pipeline    ← SHIPPED
|   Idea: Stop fighting the gateway. POST returns 202 +
|       job_id immediately. Lambda self-invokes async
|       and writes the result to a DDB table with TTL.
|       Client polls GET /<route>/status/{job_id} every
|       1.5s. Each poll is a single DDB GetItem – fast,
|       fits the 30s window with room to spare.
|   ✓ Lambda's own 60s timeout is the new ceiling, not
|       API Gateway's 30s.
|   ✓ Each piece is small and well-understood (DDB TTL,
|       self-invocation, two new GET routes).
|   ✓ The Pratt Street sign now returns in ~30-45s with
|       a stepped loading state instead of a 503.
|   ~ Cost of pivoting: ~5h. Cost of NOT pivoting: a
|       permanent excuse on every complex Melbourne sign.
```

DECISION · End-of-session signal — where should the tap target live?

Q: No-sign sessions stay open-ended forever without an explicit 'I've left'.
Where do we put the button?

```

    +-- On the home card only
    |   X Hidden when the session isn't the
    |       primary active one (e.g. multi-car)
    |
    +-- Inside session detail only
    |   X Two taps from home for the common case
    |
Tap-target location  -----+
    |
    +-- Both ← CHOSEN
        ✓ Home card: pill button below walk-back
          footer, one tap from home
        ✓ Detail: full 'End session' button,
          visible for ALL active sessions
          (incl. expiry-bearing – useful when
          leaving early and you want the PDF to
          show actual duration)
```

Commits

ffef865	feat(safety): accessibility-permit gate
84ccbb1	fix(api): retry-with-friendly-error layer for 30s API Gateway timeout (band-aid)
b325a47	feat(lambda): async-job pipeline for slow Claude routes
4b0f30a	feat(api): postJsonAndPoll helper for the async-job endpoints
0255376	feat(session): driver-signalled end-of-session
3317a96	docs: document driver-signalled end-of-session + no-sign mode

DAY 6 · Wed 21 May 2026 · Hardening + Real Domain

The day after launch turns into a quiet hardening day. Two threads in parallel — test coverage from scaffolding to green, and the proper domain migration off the personal-domain subdomain (parkproof.dsouza.tech) onto a real parkproof.com.au. Live users from yesterday's LinkedIn launch keep arriving on the legacy URL the whole time; nothing breaks.

Six commits, ~1,100 lines of test code, a deferred-by-design contract-test pivot for the lambda's prompt regressions, and an ~8-step DNS / cert / CloudFront / Cognito / CORS migration to land <https://www.parkproof.com.au> as the canonical URL. Two prompt fixes (EasyPark detection, accessibility-permit gate) caught and shipped along the way.

DECISION · URL forwarding for the apex + .au variants — registrar vs Cloudflare?

Q: parkproof.com.au (apex) + parkproof.au + www.parkproof.au all need to 301-redirect to <https://www.parkproof.com.au>. Where does the redirect live?

```
+-- A · Crazy Domains URL Forwarding
|   X $26 AUD/year PER DOMAIN. ~$52/yr for the
|   two redirecting domains. Pay a registrar
|   for what's a free feature elsewhere.
|   X Reports of HTTPS forwarding flakiness
|   (cert warnings during redirect).
|
+-- B · CloudFront alt-name + edge function
|   X Requires apex-CNAME-flattening, which
|   Crazy Domains' DNS doesn't support
|   (their validator also rejects underscore
|   prefixes in CNAME values – discovered
|   mid-cert-validation).
|
Redirect host -----+
|
+-- C · Cloudflare DNS + Page Rules <-- CHOSEN
|   / Free forever (free tier supports all needed)
|   / Real 301 with valid HTTPS
|   / CNAME flattening at apex available for free
|   if we later switch to apex-canonical
|   / Free DDoS / analytics as bonus
|   / DNS imports MX/SPF/DKIM email records
|   automatically (Titan email preserved)
|   ~ ~30 min nameserver propagation (fast)
|   ~ Manual add for ACM validation CNAMEs
|   (Cloudflare's importer skips _-prefixed)
```

DECISION · Test surface — what to cover, what to defer

Q: 110 it.todo cases across 5 risk-targeted test files. Fill all of them?

```
+-- A · Fill every case end-to-end
|     X The 18 in lambda/refresh.test.js hit the live Claude
|     API. Costs ~$0.02 per scan, slow, flaky in CI, doesn't
|     test the wiring (the prompt itself is the thing under
|     test in those scenarios).
|
+-- B · Fill only the pure-function suites (66 cases)
|     / walk-back, time-format, storage all pure logic with
|     clean edge cases. ~3-4h of focused writing, all green.
|     X Leaves visible WIP in vitest output (18 todo) on the
|     lambda file. Anti-portfolio.
|
Tests -+
|
+-- C · Fill 66 pure + write contract tests for lambda <-- CHOSEN
|     / Mock @anthropic-ai/sdk, verify request SHAPE (the prompt
|     content, model config, schema enforcement, refresh-mode
|     branching).
|     / Tests catch the regressions that ACTUALLY happen:
|     someone deleting an instruction from the prompt during
|     refactor, the model name flipping, the cache_control
|     getting dropped, etc.
|     / 92 cases green + 20 contract cases = 112/112. No WIP
|     in output. Honest about what's tested vs not.
|     / Real-user behaviour on the live site IS the integration
|     test for prompt reasoning. Don't test what you can't
|     afford to fail in CI.
```

Commits

b342dac	test: fill walk-back / time-format / storage suites (66 cases)
c62adc8	test(lambda): contract tests for refresh-mode prompt + request shape (20 cases)
a2b2b4f	fix(prompt): EasyPark / payment-methods detection regression
d9437fa	fix(prompt): ☐-only bays are parkable (with permit), not a hard block
cd6130f	feat(domain): cut over canonical URL to www.parkproof.com.au

parkproof.dsouza.tech stays alive for 7 days as fallback. Old evidence PDFs continue to verify against the public key served from either domain. Total Day 6 cost: ~\$0 on AWS (Cloudflare free tier eats the DNS work), saved ~\$52 AUD/year vs. Crazy Domains' URL-forwarding upcharge.

DAY 7 · Fri 22 May 2026 · De-identify + Web Push

A long day. ~25 commits. The brief: **strip every trace of personal identity from the public surface before posting on Reddit**, finish the Web Push pipeline end-to-end, and add the two locales the LGA-census pass had missed.

The de-identification pass alone was eight commits. The personal name was on the About-page footer, in the LinkedIn case-study link, in the melroy@ email address used as the privacy-policy contact, and — most awkwardly — in the Apple Sign-in OAuth consent screen, which read *"Continue to ParkProof using your Apple ID. Created by [Personal Name]."* Every one of those is a place a Reddit commenter could screenshot and turn into the post. The fix was methodical: swap to hello@parkproof.com.au everywhere, drop the LinkedIn refs, rotate the Apple App ID + Services ID + Key to a new bundle (au.com.parkproof.app) carrying no personal name, and re-translate the About page so the new copy stayed consistent across the 7 then-supported locales.

The Web Push pipeline shipped in four staged commits — foundation (VAPID + DDB subscription store + /push/subscribe route), service worker push + notificationclick handlers (which required switching vite-plugin-pwa from generateSW to injectManifest mode so the SW could carry our custom handler), the EventBridge Scheduler dispatcher, and a cancel-on-end-session sweep that fan-outs DeleteSchedule calls when the driver signals "I've left."

Then the visual pop refresh: AI-generated photoreal hero (Nano Banana) composited with the layered-P brand mark on the sign face, split-colour "Parking made simple." headline, purple-gradient primary CTA, dot-grid background, sage + topographic rings + corner aurora ("variant J") replacing the cool tech-grey body bg. Five style iterations on the aurora alone before the 3× radius felt right.

And finally: Indonesian + Korean. The original LGA-census pass had picked seven languages off the 2021 ABS data; a sanity-check on the actual City of Melbourne ranking surfaced Indonesian (#3) and Korean (#6) as top-non-English-spoken-at-home languages the first cut had missed. Adding them was ~2 hours of translation + flag-picker updates. Worth it for the defensibility of the locale set.

DECISION · Push schedule naming — deterministic vs random?

Q: One EventBridge Scheduler schedule per chosen reminder offset (15min, 30min, 1hr...). When the driver ends the session early, all of them need to go. How do we find them to delete?

```
+-- A · Random UUID names, ListSchedules at delete time
|
|   X Needs scheduler:ListSchedules IAM permission
|   X ListSchedules is rate-limited and paginated
|   X Race condition: a new schedule might land between
|       list and delete
|
Find ---+
|
+-- B · Deterministic names ← CHOSEN
      `parkproof-push-{session_id}-{i}`
      ✓ Cancel = 6 parallel DeleteSchedule calls, no list needed
      ✓ No new IAM permission to grant
      ✓ Idempotent on re-pick (overwrite same name)
      ✓ Trivial to reason about in logs
```

The path mattered more than the destination. Deterministic names let the cancel pathway stay as boring as `Promise.all(offsets.map(i => deleteSchedule(name(sessionId, i))))` without ever calling `ListSchedules`.

Commits (selected — 25 total)

38718ec	feat(about): drop "on the way" section + translate to 7 languages
583b7e9	feat(push): web push foundation (subscribe + persist)
f375a7a	feat(push): SW push + notificationclick via injectManifest
a97c576	feat(push): scheduler + de-identification + docs refresh
495e5f6	chore(auth): rotate Apple Bundle ID + Services ID + Key
18c5546	feat(landing): visual pop refresh — split-colour hero + gradient CTA
0eb5f2d	feat(hero): swap SVG illustration for AI-generated photoreal hero
0bf27cc	feat(bg): ship sage + topographic rings + corner aurora ("variant J")
53a7604	feat(i18n): add Indonesian + Korean (7 → 9 languages)
0756585	docs: final pre-launch consistency sweep — 9 languages, 7 days

DAY 8 · Sat 23 May 2026 · Two-App Architecture

A focused, short day — 5 commits. The biggest one is the **two-app architecture cutover**: marketing landing page at /, React PWA at /app/. Until today, the React app was serving the marketing page too — bundle bloat, slower first paint, the Router-less "home" view doing double duty as a landing page. Splitting them freed the landing to be hand-tuned static HTML/CSS (instant first paint, no JS to parse) while the PWA stayed at /app/ with all the state-machine routing intact.

The trade-off was real: every `` in the React app had to route through Vite's `BASE_URL` after the cutover or it 404'd because the asset now lived at /app/. Commit 4add63e was the cleanup pass. CloudFront also needed a viewer-request function (parkproof-uri-rewrite) to handle the trailing-slash resolution OAC + S3 REST origins don't do natively (S3 returns 404 for GET /app/; the function rewrites to /app/index.html).

The other thread was **considered-and-deferred docs**: three of them — voice confirmation (Web Speech API), localising the marketing landing, and offline / hybrid sign reading. Each got a paragraph in docs/features.md so future-me has the reasoning when the question comes back.

DECISION · One app or two?

Q: The landing page (hero + 3 feature cards + CTA) is currently the React app's "home" view. Should it stay there, or split into a static page?

```
+-- A · Keep one app, render landing as a React view
|      X Ships React + i18n + auth code to a visitor whose
|      only intent is to read the pitch and click Scan
|      X First paint blocked on bundle parse
|      X Landing redesigns require a full app redeploy
|
Apps ---+
|
+-- B · Two apps – static landing at /, PWA at /app/ ← CHOSEN
      ✓ Landing is 30KB hand-tuned HTML/CSS, instant paint
      ✓ React app stays at /app/ behind code-split chunks
      ✓ Marketing page edits ship without touching the app
      X Every absolute-path asset needs to route through
        BASE_URL – one careful cleanup pass
      X CloudFront viewer-request function needed for
        /app/ trailing-slash → /app/index.html (S3 REST
        origins don't do this natively)
```

Commits

deae35e	feat: two-app architecture — marketing landing at /, PWA at /app/
4add63e	fix: route absolute-path assets through Vite base after /app/ cutover
244db78	docs: capture two-app cutover gotchas in migration runbook
860a2fe	docs(status): mark Web Push + two-app shipped, refresh canonical
5e9e844	docs: file voice confirmation as considered-and-deferred

DAY 9 · Sun 24 May 2026 · Pre-launch Hardening (Tier 1)

The most PM-y day of the project. With the feature set frozen and CI green, I sat down on Sunday afternoon — ~48 hours before the planned Reddit launch — and explicitly enumerated **what could break under launch traffic that a green build wouldn't catch**. Six items, ~5 hours of work, framed as a "Tier 1" list: *if any of these fails on launch day, you'll cringe*. Then I worked the list.

22 commits. The single biggest catch is the one I almost shipped without noticing.

The PDF font crisis. *Item 1 on the audit list was "actually export a PDF in every supported locale." I'd never done it. The strings were translated, jsPDF was wired in, the UI worked. But jsPDF ships with Adobe Type 1 fonts that have zero glyph coverage for Devanagari, Gurmukhi, CJK, Greek, or Vietnamese. Six of nine locales rendered evidence PDFs as missing-glyph rectangles. Silent data corruption for ~30% of supported users. Would have been a "btw your evidence PDF is just boxes" Reddit comment within hours of going live.*

The fix was substantive but mechanical once the diagnosis was right: source variable-weight Noto Sans TTFs from the Google Fonts GitHub repo, write a Python build-time script that subsets each font down to just the glyphs that appear in the locale JSONs (94% size reduction — Chinese Simplified went from 17MB to 893KB), self-host the subsets on CloudFront, lazy-load the right one per locale at PDF export time. Two hours of focused work. Ships clean.

The interesting bit isn't the fix — it's that the *audit* found it. A different sequencing (skip the audit, trust the build, hit publish) would have shipped the bug.

DECISION · The Tier 1 audit list

Q: Feature work is done, CI is green, the site is technically launch-ready.
What could still cringe on launch day?

The list I wrote on Sunday afternoon:

1. PDF export in every locale, not just English
→ Caught: 6/9 locales rendered as glyph boxes. Fixed.
2. Reddit / iMessage / LinkedIn share preview
→ Caught: no OG tags on the new marketing landing. Fixed.
3. First Reddit visitor's cold-start latency
→ Caught: 1-3s p99 cold-start. Fixed with EventBridge warmer pinging every 5 min.
4. Reminder visibility after setting
→ Caught: pushes were invisible after configuration. Built self-service management surface.
5. CloudWatch dashboard for the launch window
→ Built. 9 widgets across 4 rows.
6. Rollback playbook in CLAUDE.md
→ Written. Symptom triage table + revert recipes.

Every item is a thing that doesn't fail a test, can't be caught by lint, and isn't worth a feature PR — but each one, left unchecked, becomes a launch-day cringe moment.

DECISION · PDF font sourcing strategy

Q: jsPDF needs a TTF (not WOFF/WOFF2) registered per locale at export time. Six locales need non-Latin scripts. What's the smallest, most maintainable pipeline?

```
+-- A · @fontsource v5 from npm
|   X Ships WOFF only. jsPDF needs TTF.
|   X Adds 5 new top-level dependencies for what is
|       fundamentally a static asset problem.
|
+-- B · Bundle full Noto Sans TTFs with the SPA
|   X Chinese Simplified is 17MB uncompressed.
|       Korean is 11MB. Inflates the bundle ~50MB for
|       users who'll never trigger PDF export in those
|       locales.
|
Fonts --+
|
+-- C · Subset + self-host + lazy-load ← CHOSEN
|   • Source: google/fonts GitHub raw URLs
|   • Build: scripts/_subset_pdf_fonts.py runs
|       pyftsubset over each locale JSON's unique
|       glyphs, --no-hinting --desubroutinize
|   • Result: 94% size reduction (CJK 17MB → 893KB)
|   • Runtime: src/lib/pdf-fonts.ts lazy-fetches
|       the right subset for the active locale
|   ✓ Bundle stays small
|   ✓ Subset is reproducible – re-run the script
|       when locales change
|   ✓ jsPDF gets a real TTF with the glyphs it
|       needs and nothing else
```

DECISION · Reminder visibility — invisible vs managed

Q: After picking reminder offsets, the schedules live in EventBridge – invisible from the app. The .ics event renders in the user's calendar, but Web Push is a black box: set-and-forget, no way to see what's queued, no way to cancel without redoing the whole session. That violates the trust principle the rest of the product is built on.

```
+-- A · Don't show anything; trust EventBridge
|   X User can't verify reminders were set
|   X "I never got the push" – no way to tell from
|       the app whether it was queued, denied, or SW-failed
|
+-- B · Show a passive list – no controls
|   X Half-measure. "3 reminders queued" but still
|       can't change them.
|
Vis  ---+
|
+-- C · Full self-service management ← CHOSEN
|   • Lists every queued fire_at, time-formatted
|   • Per-row × cancel (DeleteSchedule one-shot)
|   • "+ Add reminder" picker filtered to offsets
|       that haven't already fired or aren't queued
|   • Preserves per-reminder body text across edits
|       (bug caught and fixed pre-launch)
|   ✓ Went from fire-and-forget to managed-and-trusted
|       in 2.5 hours
```

Commits (selected — 22 total)

bfc3d24	docs+i18n : soften legal-promising language across all surfaces
9fa6ae6	feat(landing) : public /verify/ page with openssl walkthrough
eaad3d16	feat(verify) : multi-lingual /verify pages in all 9 locales
d68cd07	fix(i18n) : include querystring in detection chain
5eda011	fix(pdf) : render non-Latin scripts via self-hosted Noto Sans
5e782ba	feat(lambda) : pre-warm via EventBridge to kill cold-start
0efe3d4	docs : add rollback playbook to CLAUDE.md
6307c64	feat(seo) : OG cards + canonical + robots/sitemap on landing
732d584	feat(pwa) : service worker update banner
937958d	feat(ops) : launch-day CloudWatch dashboard + 30-day log retention
7706116	feat(reminders) : scheduled-reminder visibility + per-session mgmt
62146aa	feat(a11y) : reduced-motion respect + OAuth callback splash
c4e3e84	feat(polish) : Tier 3 polish bundle — 5 self-contained launches

DAY 10 · Mon 25 May 2026 · Launch Eve

Three commits, three categories of fix: a genuine UX gap, a stale-asset refresh, and Reddit-thumbnail polish. Tomorrow morning is the launch.

The gap I almost shipped: **first-time visitors who already had an account on another device had no path to sign in**. The first-time landing surfaced the gradient "Scan a parking sign" CTA and nothing else. Returning users see auth options in the home view's overflow menu, but a first-time visitor on a new device — say, a returning Mac user opening the iOS app fresh — had to *create a dummy session* to discover the sign-in path. That's a fatal cross-device-recovery gap. The fix was small: pass an `onSignInCta` prop from `App.tsx` to `LandingFeatures` (only when `auth.configured && !auth.user`), which renders a white-on-paper secondary button directly below the gradient CTA. Reuses the existing `home.signInToSync` i18n key, so no translation churn across 9 locales.

The first attempt at this fix put a small underlined link at the *bottom* of the landing — "Already have an account? Sign in." Feedback in review was decisive: *"I don't like where it's located. It's genuinely hidden at the bottom of the screen. I think it should be a button below 'Check a parking sign'."* Moved it. Second attempt shipped. The dialogue mattered more than the diff: it pushed the sign-in path from *findable-if-you-look* to *unmissable-on-arrival*.

Remaining work: a screenshot refresh (the demo grid now reflects the secondary sign-in button) and a Reddit-thumbnail tweak — the "ParkProof" wordmark on the carousel slide read too small at Reddit feed size, so a brand-blue header strip with the word in large Georgia Bold white got composited on top. Tiny edit, proportionally large impact on first-impression brand recognition.

Then case-study tightening: ~260 words trimmed from 3,603 to 3,344 on seven targeted cuts — hedge sentences in the Problem section, walk-back disclaimers, the "hiring PM would recognise" sentence that read as fishing, a screenshot-pipeline hindsight bullet that didn't earn its weight against the others, and a stack-summary section that was 90% there but reading slightly verbose.

DECISION · Sign-in CTA placement on the landing

Q: First-time visitors with no session see only the gradient "Scan a parking sign" CTA. Cross-device-recovery users (account exists, new device) have no path to sign in. Where does the sign-in button go?

```
+-- A · Bottom-of-page text link
|     X Hidden below the fold on iPhone widths
|     X Reads as low-importance footer fine print
|     X Explicitly rejected in review
|
+-- B · Top-right of the header, next to language picker
|     X Crowds the header in narrow viewports
|     X "Sign in" pattern is button-style, not
|       icon-button-style – wouldn't read naturally in
|       a 32px corner slot
|
Slot ---+
|
+-- C · Secondary button directly below scan CTA ← CHOSEN
      • White-on-paper, smaller padding than gradient
      • Renders ONLY when auth.configured && !auth.user
      • Reuses home.signInToSync i18n key across all
        9 locales – no translation churn
      ✓ Unmissable on arrival
      ✓ Visual hierarchy preserved – gradient still
        wins the eye
      ✓ Two-line tap surface, mobile-thumb friendly
```

Commits

23c49c8	fix(auth): expose sign-in path for first-time visitors
3083867	fix(auth): move sign-in CTA next to the primary scan button
cbc886a	docs: regenerate screenshots to reflect today's UI changes

What this build proved

1 · Sequencing > stack

Tier-A features shipped before the auth pivot. The home countdown shipped before i18n. The screenshot harness shipped before the README rewrite. Order matters more than tooling — every later piece either built on or was made cheaper by the order of what came before.

2 · Anonymous-by-default is a real design principle

Every feature works without a login. Cloud sync is opt-in, never gated. The local copy stays the source of truth even when signed in. This isn't just a UX choice — it's why the safety gates (paid-parking, accessibility-permit) and the no-sign flow could ship without an auth dependency, and why the architecture stayed simple enough that the Day 5 timeout pivot didn't have to fight Cognito's session lifecycle on top of everything else.

3 · The right pivot is the one that simplifies the system

Two architectures failed on Day 5. Both would have required deep knowledge of AWS-account-level quirks to debug. The third — async polling — required none. It decomposes into three boringly small pieces (a TTL'd DDB table, an `InvocationType: 'Event'` self-invoke, two new GET routes), all of which fit in their respective service's normal envelope. The pivot wasn't more clever; it was less clever, and that was the point.

4 · Build journals are useful in the same week

Most build retrospectives are written months later, when the dead ends have been softened into a tidy success story. The 5 hours of Day 5 dead-ends and the silent PDF-font bug from Day 9 are still raw in this document because the document is being written hours after they happened. That rawness is the value — a future hiring manager (or future-me planning the next build) gets to see the trade-offs as they actually were.

5 · The audit is the deliverable

The single most valuable session of these ten days was the 5-hour Tier-1 pre-launch audit on Sunday afternoon. Naming six things that *don't fail tests but do fail on launch day*, then working the list, caught the PDF-font crisis that would have shipped a silent data-corruption bug to ~30% of supported users. The audit cost five hours; the upside is not having to retract a Reddit post.

6 · De-identification is its own feature

Going from *"functional product with my name on it"* to *"functional product that could be anyone's"* took a focused half-day. Surfaces that needed sweeping: About-page footer, privacy contact email, Apple Sign-in consent screen, README/case-study LinkedIn references, VAPID `mailto:` subject, source-code author tags. Most of these are invisible until they're embarrassing. Worth doing once, on purpose, in one pass.

7 · Trust principles compound or compound-rot

The reminder-visibility gap on Day 9 was a small thing — *"I just told the app to set 3 push reminders; where are they?"* — but it violated the trust principle the rest of the product is built on (anonymous by default, transparent about what's running, user is in control). Fixing it was 2.5 hours. Not fixing it

would have rotted the trust principle by one notch, and the next gap would have rotted it by another. These don't add linearly.

Appendix · Full commit log

Every commit on main, oldest first.

#	Hash	Date	Message
1	98bd332	16 May	chore: initial commit
2	e6396b4	16 May	feat(reminders): multi-select offsets with auto-disabled past chips
3	5a236b4	16 May	feat(tooling): automated screenshot harness via Playwright
4	98d2f17	16 May	fix(pdf): surface export errors, guard malformed signatures, fix expired-session copy
5	a58cd4c	17 May	feat(home): live 'Currently parked' card with urgency-coloured countdown
6	9c0691b	17 May	feat(time): date-aware 'Until' / 'Move by' / reminder labels
7	402f531	17 May	feat+fix: Tier-A features + three gotcha fixes
8	1d20352	17 May	docs(readme): refresh stale sections after Tier-A + gotcha shipments
9	c393511	17 May	feat(auth+sync): Cognito sign-in, DynamoDB + S3 cloud sync, end-to-end
10	c8e7391	17 May	fix(auth): SPA boot polyfill, CORS for GET + Authorization, export as PDF
11	a5cd755	17 May	docs(readme): align with the now-shipped auth + PDF-export reality
12	3127e86	17 May	docs(readme): flag pending Apple federation
13	84c7d99	17 May	feat(i18n): five-language support with flag selector
14	d5f6f3b	17 May	docs(readme): align with multi-lingual + auth + cloud-sync reality
15	9542071	17 May	fix(i18n): restore inline emphasis lost when strings were flattened
16	d3d0419	17 May	fix: ReuseCard Italian half-translation + stale-chunk PDF export crash
17	6e84cff	17 May	feat(i18n): Hindi + Punjabi + PDF translations + dropdown selector
18	2cdc257	17 May	docs(readme): Apple federation live
19	03f8006	17 May	feat(telemetry): AI feedback Layer 2
20	3ab2e6f	18 May	feat(domain): cut over to parkproof.dsouza.tech
21	79d5acc	18 May	chore(public): drop unused / off-brand orphan assets
22	fb1fa8b	18 May	docs: refresh README + CLAUDE for current shipped surface
23	7bbe469	18 May	chore: add MIT LICENSE
24	34cef80	18 May	docs(readme): fix stale numbers + add cloud-sync to feature list
25	8b6d103	18 May	fix(i18n): home-screen cards drift off-centre when translations wrap
26	5116c7e	18 May	feat(screenshots): expand pipeline to 16 captures + add real fixtures
27	558aed1	18 May	docs(screenshots): regenerate all 16
28	73df423	18 May	docs(readme): restructure Demo into 4 themed sub-grids
29	e8388b9	18 May	docs: add PM case study
30	b82eef6	18 May	fix(home): '+N more' pill is now a real tap target
31	e3e1ceb	18 May	fix(sync): cross-device images now actually load
32	9fb2d45	19 May	docs: pre-launch audit fixes
33	11dcad4	19 May	feat: paid-parking gate + no-sign-here flow
34	ffef865	20 May	feat(safety): accessibility-permit gate
35	84ccbb1	20 May	fix(api): retry-with-friendly-error layer for 30s API Gateway timeout
36	b325a47	20 May	feat(lambda): async-job pipeline for slow Claude routes
37	4b0f30a	20 May	feat(api): postJsonAndPoll helper for the async-job endpoints
38	0255376	20 May	feat(session): driver-signalled end-of-session
39	3317a96	20 May	docs: document driver-signalled end-of-session + no-sign mode

40	b342dac	21 May	test: fill walk-back / time-format / storage suites (66 cases)
41	c62adc8	21 May	test(lambda): contract tests for refresh-mode prompt + request shape
42	a2b2b4f	21 May	fix(prompt): EasyPark / payment-methods detection regression
43	d9437fa	21 May	fix(prompt): ■-only bays are parkable (with permit), not a hard block
44	cd6130f	21 May	feat(domain): cut over canonical URL to www.parkproof.com.au
45	38718ec	22 May	feat(about): drop 'on the way' section + translate to 7 languages
46	61b3e1e	22 May	fix(about): strip personal identity from in-app footer
47	583b7e9	22 May	feat(push): web push foundation (subscribe + persist)
48	62ae67d	22 May	fix(push): add @aws-sdk dynamo packages at root for test script
49	f375a7a	22 May	feat(push): SW push + notificationclick via injectManifest
50	a1f8b4a	22 May	fix(push): check actual subscription, not just permission
51	a97c576	22 May	feat(push): scheduler + de-identification + docs refresh
52	848bd44	22 May	docs(readme): finish the audit — file tree + script table
53	eea6bdb	22 May	docs(screenshots): regenerate all 16 with current UI
54	495e5f6	22 May	chore(auth): rotate Apple Bundle ID + Services ID + Key
55	b8c0532	22 May	feat(permit-zone): acknowledgment gate for permit holders
56	18c5546	22 May	feat(landing): visual pop refresh — split-colour hero + gradient CTA
57	3296efa	22 May	style(polish): gradient CTAs across screens + dot-grid background
58	50fcaeb	22 May	feat(reminders): open-ended picker for no-sign sessions
59	cf50f36	22 May	style(bg): warm paper background (#FAF6EE) instead of tech-grey
60	d1a30af	22 May	style(hero): dimensional hero illustration upgrade
61	0eb5f2d	22 May	feat(hero): swap SVG illustration for AI-generated photoreal hero
62	b0aa621	22 May	fix(hero): re-bake composite — brand mark was 63px outside sign
63	681e50b	22 May	fix(hero): App.tsx returning-user header still pointed at deleted .svg
64	7eaac63	22 May	feat(landing): promote hero photo to full-width banner
65	bc65ef7	22 May	refactor(hero-bake): promote sign-bbox detection into the script
66	a9bcb61	22 May	feat(hero): swap in higher-res Nano Banana photo (1152x928)
67	28b5847	22 May	fix(i18n): localise countdown 'left' label + swap car-photo fixture
68	298ed17	22 May	fix(screenshots): screenshot 07 was reading seeded fixture, not live
69	0bf27cc	22 May	feat(bg): ship sage + topographic rings + corner aurora ('variant J')
70	1aba9f9	22 May	style(bg): bump aurora from 500px to 2000px (4× linear)
71	90ccae4	22 May	style(bg): dial aurora back from 4× to 2× (sweet spot)
72	778bf28	22 May	style(bg): aurora to 3× — Goldilocks middle ground
73	53a7604	22 May	feat(i18n): add Indonesian + Korean (7 → 9 languages)
74	0756585	22 May	docs: final pre-launch consistency sweep — 9 languages, 7 days
75	deae35e	23 May	feat: two-app architecture — marketing landing at /, PWA at /app/
76	4add63e	23 May	fix: route absolute-path assets through Vite base after /app/ cutover
77	244db78	23 May	docs: capture two-app cutover gotchas in migration runbook
78	860a2fe	23 May	docs(status): mark Web Push + two-app shipped, refresh canonical
79	5e9e844	23 May	docs: file voice confirmation as considered-and-deferred
80	bfc3d24	24 May	docs+i18n: soften legal-promising language across all surfaces
81	1a4eb51	24 May	docs: regenerate screenshots + PDFs after tamper-proof sweep
82	d870c9a	24 May	fix(landing): drop EVIDENCE.PDF tab on Evidence section image
83	39f6b33	24 May	fix(landing): swap PDF mockup's Sign photo from scanner to real sign

84	411e536	24 May	docs: file offline / hybrid sign reading as considered-and-deferred
85	9fa6ae6	24 May	feat(landing): public /verify/ page with openssl walkthrough
86	5ebe6b3	24 May	feat(privacy): federated sign-in section in in-app privacy policy
87	5fc42a3	24 May	fix(wiring): PEM at site root + /verify discoverability
88	eea3d16	24 May	feat(verify): multi-lingual /verify pages in all 9 locales
89	b68b9cd	24 May	feat(verify): auto-detect chip on /verify/ + locale-aware link
90	210a580	24 May	refactor(about): translated, contextual push-subscribe block
91	d68cd07	24 May	fix(i18n): include querystring in detection chain
92	c243877	24 May	fix(landing): pass ?lng= to chip target URL
93	4c6bc4f	24 May	docs: file landing translation as considered-and-deferred
94	5eda011	24 May	fix(pdf): render non-Latin scripts via self-hosted Noto Sans
95	5e782ba	24 May	feat(lambda): pre-warm via EventBridge to kill cold-start
96	0efe3d4	24 May	docs: add rollback playbook to CLAUDE.md
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102	19cff0a	24 May	docs: refresh case study + README + lessons for launch readiness
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107	cbc886a	25 May	docs: regenerate screenshots to reflect today's UI changes